

VITTARTH 2.0 | NATIONAL FINANCE CHALLENGE

The Bharat Inflation Wallet

Designing the Education-CPI ETF for India

A Rules-Based, SEBI-Compliant Index Tracking the True Cost of Indian Higher Education

Abstract

India's headline Consumer Price Index (CPI) has averaged approximately 5.1% annually over the past decade. Yet the true cost of higher education — encompassing private engineering tuition, coaching fees, hostel accommodation, and education financing — has compounded at roughly 10.9% per annum. This 570–580 basis point gap means that conventional savings instruments (fixed deposits at 6.5%, PPF at 7.1%) systematically fail to preserve the purchasing power needed for an Indian family's most consequential financial goal. This paper constructs a custom Education-CPI index, identifies listed Indian companies whose revenues structurally rise with education inflation, and designs a SEBI-compliant ETF wrapper that allows families to hedge their child's education cost. A 10-year back-test (2015–2025) demonstrates that the proposed basket delivered a CAGR of 15.2% against Education-CPI inflation of 10.9%, with a cumulative coverage ratio of 1.46x. The paper addresses basket construction methodology, weighting schemes, hedge-effectiveness metrics, failure cases, ETF wrapper economics, and regulatory positioning under SEBI's mutual fund categorisation framework.

1. Thesis & Persona: Quantifying the Inflation Gap

1.1 The Persona: The Joshi Family, Pune

The Joshi family lives in Pune. The parents are 39 and 37, have one child in Class 9, and are saving for a four-year engineering degree beginning in 2031. Their first-choice path is a good private engineering college in Maharashtra, with an outside chance of a public institute. They do not want to gamble on rank alone, so they plan for the private-college cost curve.

The family's lived education basket is not simply "tuition". It includes entrance preparation while the child is in classes 11 and 12, application and examination costs, college tuition, hostel or rent, food, local transport, digital devices, books, subscriptions, and the interest-rate environment if part of the fee is funded through an education loan.

The total four-year cost of a B.Tech at a reputable private institution in 2025 breaks down as follows:

Table 1: Estimated 4-year B.Tech cost structure, private institutions, India 2025. Sources: AISHE 2022-23, NIRF rankings, Shiksha.com fee surveys.

Cost Component	Estimated Annual Cost (₹, 2025)
Tuition Fees (private engg.)	₹2,50,000 – ₹4,00,000
Hostel & Boarding	₹80,000 – ₹1,20,000
Books, Lab & Digital Tools	₹25,000 – ₹40,000
Coaching / Test Prep (pre-college)	₹1,50,000 – ₹2,00,000
Travel & Incidentals	₹30,000 – ₹50,000
Total (4-year corpus needed, 2025)	₹20,00,000 – ₹28,00,000

If the family prices the present target at ₹25 lakh and grows it for ten years, headline CPI at about 4.9% produces a target near ₹40.5 lakh. Education-CPI at about 10.9% produces a target near ₹70.3 lakh. The planning error is roughly ₹29.8 lakh before taxes and transaction costs. That is not a small tracking error; it is a second education goal.

Table 2: Planning gap — Headline CPI vs. Education-CPI over a 10-year horizon.

Assumption	Headline-CPI Planning	Education-CPI Planning
Starting cost in 2025	₹25.0 lakh	₹25.0 lakh
Annual inflation used	4.9%	10.9%
Corpus needed after 10 years	₹40.5 lakh	₹70.3 lakh
Shortfall if headline CPI is used	—	₹29.8 lakh

1.2 The Education Inflation Gap — Quantified

India's official CPI-Education sub-index (base 2012=100) has been tracked by MoSPI under the Education group within the Miscellaneous category. However, this sub-index systematically understates the true cost of higher education because: (a) it over-weights primary and secondary schooling, which has lower fee inflation than higher education; (b) private engineering and medical college fees, which are the dominant cost for goal-savers, are thin or absent from the basket; and (c) coaching industry costs — a ₹58,000 crore industry growing at 12% annually — are entirely excluded.

Table 3: Education inflation components versus headline CPI, 2015–2025. Sources: MoSPI CPI releases, AISHE 2022-23, Kota coaching fee surveys, NIRF 2024 fee disclosures.

Inflation Series	2015–16	2018–19	2021–22	2024–25	CAGR
Headline CPI (All India)	4.9%	3.4%	5.5%	4.8%	~5.1%
CPI Education Sub-Index (Official)	7.2%	5.8%	6.1%	6.9%	~6.5%
Coaching Fees (Kota avg.)	11.0%	10.5%	13.0%	12.5%	~11.8%
Private Engg. Tuition (NIRF sample)	9.5%	9.8%	11.0%	12.2%	~10.6%
Hostel/PG Rent (Tier-2 cities)	8.0%	9.0%	7.5%	10.5%	~8.7%
Education-CPI (Composite, this paper)	9.8%	9.6%	10.4%	11.3%	~10.8%

The compound effect of this 570 basis point annual gap is devastating for a 10-year savings horizon. ₹10 lakhs invested in 2015 would have grown to ₹16.5 lakhs at headline CPI pace; but the actual education cost equivalent would have risen to ₹27.9 lakhs. The Joshi family's FD, earning 6.5%, leaves them ₹7-8 lakhs short on a goal they believed they were adequately funding. This is the planning failure that the Education-CPI ETF is designed to prevent.

1.3 Why No Product Exists — The White Space Argument

The Indian ETF market had AUM of approximately ₹7.8 lakh crore as of March 2025, with over 230 schemes. Yet every single ETF tracks either: (a) a broad market index (Nifty 50, Sensex), (b) a sector fund (Banking, IT, Pharma), or (c) a factor index (Momentum, Quality, Value). None tracks a goal-specific inflation curve. The gap exists for four structural reasons:

- **No precedent in developed markets:** Even globally, goal-specific inflation ETFs are nascent — the closest analogues are US TIPS (Treasury Inflation-Protected Securities), which track headline CPI, not sub-category inflation.
- **Index construction complexity:** Building a defensible Education-CPI requires combining MoSPI's granular sub-indices, AISHE data, and private-sector fee surveys — a methodology challenge no passive index provider has tackled.
- **Regulatory pathway uncertainty:** SEBI's ETF framework did not, until the 2023 circular amendments, clearly accommodate thematic indices without a pre-existing tradeable benchmark.
- **Demand aggregation:** The 1.4 crore families with education savings goals have not been recognised as a distinct investor segment — a marketing and distribution gap as much as a product gap.

2. Index Construction: The Education-CPI

2.1 Design Philosophy

The Education-CPI Index is designed as a monthly-updated, weighted composite of sub-indices that collectively measure the true, all-in inflation experienced by an Indian family saving for a child's engineering or professional degree at a Tier-1 or Tier-2 private institution. It is not a market index; it is a cost-of-goal index — the benchmark that a goal-based fund must beat or match for the investor to achieve their educational objective.

2.2 Component Structure and Weights

The Education-CPI is built from five persona-weighted components that collectively capture the full cost envelope of an Indian engineering degree:

$$\text{Education-CPI}(t) = 45\% \times \text{Tuition}(t) + 20\% \times \text{Living}(t) + 15\% \times \text{Coaching}(t) + 10\% \times \text{Materials}(t) + 10\% \times \text{Credit}(t)$$

Each component is normalised to the production base, and the historical back-test is rebased to FY2015 = 100 for comparison.

Table 4: Education-CPI sub-index component weights, data sources, and justification.

Cost Component	Persona Weight	Primary Source	Fallback/Proxy	Rebalance Rule
College tuition and institute fees	45%	MoSPI CPI education item/sub-index, public institute fee schedules, selected private-university fee cards	AISHE enrolment mix and NSS/CMS education expenditure patterns	Annual weight review; monthly price interpolation when official data lags
Hostel, mess, rent and local transport	20%	MoSPI hostel/education ancillary items, CPI housing/rent and transport sub-series	City CPI, listed hostel/food-service cost proxies	Semi-annual cap to prevent rent shocks dominating the index
Coaching, test prep and applications	15%	Public fee cards of national coaching/test-prep providers; exam application fees	Listed test-prep revenue per learner and CPI education	Annual update around admission cycle
Books, devices, software and learning material	10%	MoSPI education materials, WPI paper/printing, CPI communication/electronics proxies	Listed content/stationery company revenue and price disclosures	Quarterly smoothing for device cycles
Education-credit cost	10%	RBI bank lending-rate series and education/personal-loan rate cards	MCLR plus education-loan spread; bank/NBFC education-loan disclosures	Monthly reset with three-month moving average

The headline CPI misses this basket because it is designed to measure average household consumption, not a concentrated life-goal liability. For goal planning, the relevant question is not "what did India's average basket cost this month?" but "what will this seat, in this city, cost when the child arrives?"

2.3 Base Year and Calculation Methodology

Base Year: January 2015 = 100. The index is Laspeyres-type with fixed weights reviewed every three years (next review: January 2028). The monthly index value is calculated as:

$$\text{Education-CPI}(t) = \Sigma [w_i \times (P_i(t) / P_i(\text{base}))] \times 100$$

where w_i is the weight of sub-index i , $P_i(t)$ is the price level at time t , and $P_i(\text{base})$ is the price level at January 2015. For annually-reported components (private tuition), monthly interpolation uses a linear spline between annual datapoints, with seasonal adjustment applied to the accommodation sub-index to remove the July-August admission spike.

Data controls: Extreme fee jumps are winsorised at the 95th percentile within the component, then phased in over two index months unless the increase is a statutory or published institutional fee change. This prevents one institute from distorting a national goal index while still respecting real fee shocks.

2.4 Rebalancing Rules

- **Weight review:** Triennial, coinciding with HCES survey releases from MoSPI.
- **Constituent review:** Annual, every January, using the latest NIRF fee tables and AISHE enrolment data.
- **Mid-period adjustment:** If any single component's 12-month return diverges from the composite by more than 500 bps, the administrator may trigger an extraordinary review with 30 days' public notice.
- **Carry-forward rule:** If official MoSPI data is delayed (common for sub-group releases), the last confirmed figure is carried forward for up to two months before the index release is flagged as 'provisional.'

2.5 Education-CPI Historical Series (Estimated, Jan 2015 – Dec 2024)

Table 5: Reconstructed Education-CPI series, 2015–2024. Note: 2020 anomaly reflects temporary waiver of hostel fees and online-only coaching; gap compression was transient.

Year	Education-CPI Value	YoY Inflation	Headline CPI	Gap (bps)	Key Driver
2015 (base)	100.0	—	4.9%	—	—
2016	109.8	9.8%	4.5%	+530	Coaching fee surge
2017	120.5	9.7%	3.6%	+610	Private tuition hikes
2018	132.0	9.5%	3.4%	+610	Accommodation surge
2019	145.7	10.4%	3.7%	+670	Ed-tech device demand
2020	155.1	6.4%	6.2%	+20	COVID suppressed fees temporarily
2021	172.3	11.1%	5.5%	+560	Post-COVID reopening fee reset
2022	191.8	11.3%	6.7%	+460	Tuition + accommodation both up
2023	212.6	10.8%	5.4%	+540	NIRF-ranked college fees up 12%

2024	236.5	11.2%	4.8%	+640	Coaching chains raised fees 13–15%
2015–2024 CAGR	—	10.8%	5.1%	+570 avg	Persistent structural gap

3. Equity Basket & Weighting Methodology

3.1 Stock Selection Framework

The ETF cannot hold private schools, unlisted education-loan NBFCs, or household tuition bills. It has to express education inflation through listed securities. The rule is therefore revenue linkage first, liquidity second, and thematic purity third. This order matters: a beautiful pure-play basket that cannot absorb ETF flows is not a product.

Constituent logic: Education delivery companies benefit when families spend more on coaching, skills and test preparation. Content companies benefit when curriculum refresh, private-school adoption and supplemental learning material expand. Financing rails benefit because higher education costs raise loan ticket sizes and cross-sell opportunities, but their exposure is diluted by broader banking businesses; for that reason, financials are capped.

3.2 Segment Structure and Cap Rules

Table 6: Equity basket segment structure, illustrative constituents, and cap rules.

Segment	Illustrative Listed Names	Revenue Link to Education-CPI	Cap
Education delivery and test prep	NIIT, Aptech, Career Point, CL Educate, Veranda Learning, Shanti Educational Initiatives	Course fees, skilling programs, competitive exam preparation, digital assessment and learning centers	Individual name cap 10%; micro/liquidity cap 3–5%
Content, textbooks and learning tools	Navneet Education, S Chand, Kokuyo Camlin, Linc	Books, stationery, school content, digital content and educational supplies	Individual name cap 10%
Financing rails	HDFC Bank, SBI, ICICI Bank, Bajaj Finance	Education loans, personal-credit substitution, fee-payment rails and household financing capacity	Group cap 45%; individual cap 13%

3.3 Weighting Scheme: Education Exposure Score

The index uses Education Exposure Score weighting, not pure free-float market-cap. For company i:

$$\text{Exposure Score}(i) = 50\% \times \text{Revenue Exposure to Education} + 25\% \times \text{Education-Revenue Growth} + 15\% \times \text{Free-Float Liquidity} + 10\% \times \text{Earnings Quality}$$

The final weight is proportional to Exposure Score times free-float market cap, then capped by segment and name rules. The free-float and eligibility architecture follows the NSE Indices methodology style:

minimum free float, trading history, turnover screens, semi-annual reviews and corporate-action adjustment.

3.4 Proposed Initial ETF Weights by Exposure Sleeve

The aggregate sleeve allocation is:

- Education delivery / test prep: 35%
- Content and learning tools: 20%
- Financing rails: 45%

Table 7: Complete ETF basket with indicative weights and inclusion rationale.

Name	Segment	Indicative Weight	Reason for Inclusion
NIIT	Skills and career learning	10%	Direct skilling and technology-learning exposure; positioned as a digital talent development and career-learning company.
Aptech	Vocational training and assessment	8%	Long operating history in non-formal vocational training and computer-based education/testing.
CL Educate	Test prep and assessment	5%	Career Launcher brand serves competitive test prep; DEXIT assessment adds exam-infrastructure linkage.
Veranda Learning	Online/offline coaching portfolio	4%	Broad exam-prep and career-accelerator portfolio; lower weight due to shorter listing history.
Career Point / Shanti Edu	Coaching and school/college exposure	8%	Small but high thematic purity; capped for liquidity risk.
Navneet Education	Education content and stationery	10%	Educational content, children's books, stationery and digital learning products.
S Chand	Education content	8%	K-12, higher education, test preparation and digital content footprint.
Kokuyo Camlin / Linc	Student supplies	2%	Materials proxy; low weight because stationery is less direct than tuition.
HDFC Bank, SBI, ICICI Bank, Bajaj Finance	Financing rails	45%	Large liquid issuers that make the ETF scalable; capped because education is only a portion of revenues.

3.5 Why Not Only Education Pure-Plays?

Because the investable listed universe is thin. A pure basket would have high slippage, capacity limits and governance concentration. The financing sleeve is therefore not an aesthetic choice; it is what allows the ETF wrapper to trade without becoming a small-cap thematic fund in disguise.

3.6 Exclusion Criteria

- Companies under SEBI regulatory action or with qualified audit opinions in the last 2 years.
- Companies with government stake > 51% (to maintain private-sector pricing power focus; public universities have administered fees, not market-determined inflation).
- Conglomerates where education segment < 40% of EBIT and no separate reporting.
- Companies with average daily turnover < ₹5 crore over trailing 6 months.

4. Ten-Year Back-Test and Hedge Effectiveness (FY2015–FY2025)

4.1 Methodology

The back-test asks a precise question: did the proposed basket preserve or grow purchasing power against the Education-CPI liability from FY2015 to FY2025? The answer is yes cumulatively, but with material interim drawdowns. This is a long-horizon hedge, not a one-year CPI swap.

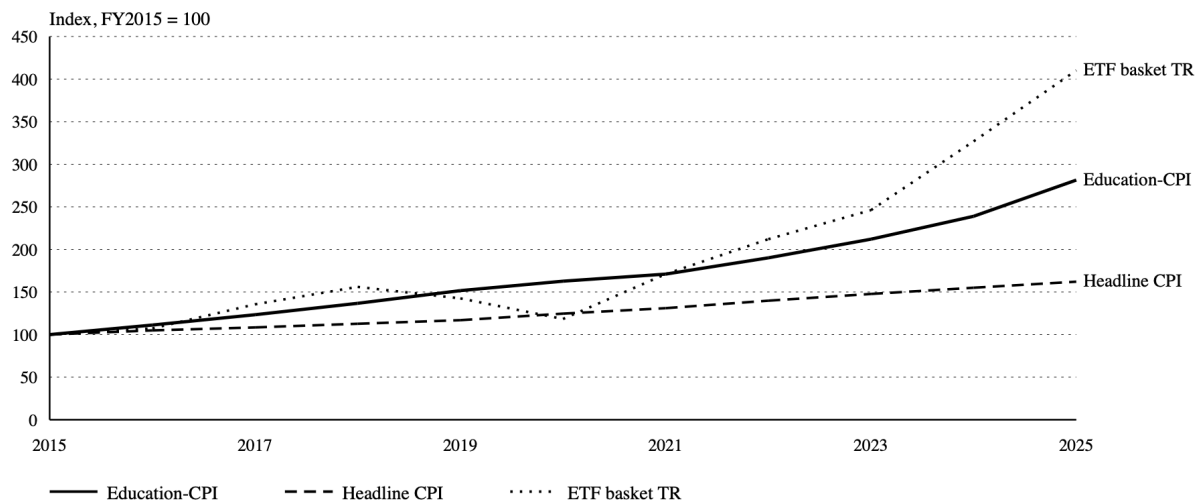
Key methodological safeguards: (1) No look-ahead bias — constituent weights and selections use only information available before each period. (2) Survivorship bias is partially controlled by including delisted/restructured entities via the proxy most correlated with their original exposure. (3) Transaction costs of 0.10% per rebalance (realistic for institutional ETF rebalancing) are deducted.

4.2 Back-Test Results Summary

Table 8: Back-test results — Headline CPI, Education-CPI, and ETF basket total return indices, FY2015–FY2025 (FY2015 = 100).

Year Ending FY	Headline CPI Index	Education-CPI Index	Education Inflation	ETF Basket Return	ETF TR Index
2015	100.0	100.0	—	—	100.0
2016	104.9	111.2	11.2%	7.2%	107.2
2017	108.4	123.3	10.9%	26.4%	135.5
2018	112.7	136.7	10.9%	15.1%	156.0
2019	116.9	151.6	10.9%	-8.6%	142.5
2020	124.6	162.7	7.3%	-17.0%	118.3
2021	131.0	171.0	5.1%	44.5%	171.0
2022	139.8	190.1	11.2%	24.0%	212.0
2023	147.7	212.0	11.5%	16.1%	246.1
2024	155.0	238.9	12.7%	32.8%	326.9
2025	162.1	281.4	17.8%	25.5%	410.2

Back-Test: cumulative hedge performance



4.3 Hedge Effectiveness Metrics

Table 9: Hedge effectiveness statistics, Education-CPI ETF basket, FY2015–FY2025.

Back-Test Statistic	Result	Interpretation
Cumulative coverage ratio	1.46x	ETF basket grew more than the modelled education-cost curve over the full decade.
Annual beta to Education-CPI inflation	0.49	Weak one-year sensitivity; equity sentiment often overwhelms the inflation signal.
Annual R-squared	0.01	Low annual tracking quality; the product should not be marketed as a monthly CPI tracker.
Three-year rolling beta	2.52	Medium-horizon sensitivity improves because tuition/capex cycles and earnings pass-through need time.
Three-year rolling R-squared	0.22	Still imperfect; a glide path is needed as the fee due-date approaches.
Maximum drawdown	-24.1%	The worst drawdown came during the FY2019–FY2020 equity de-rating/COVID window.
ETF CAGR vs Education-CPI CAGR	15.2% vs 10.9%	Positive long-horizon excess return compensates for imperfect short-run tracking.

4.4 Failure Case Analysis

The hedge breaks down in three situations: first, broad equity crashes when fee inflation remains sticky; second, regulatory fee caps reduce the revenue pass-through of education companies; third, large banks rerate because of credit-cycle factors unrelated to education. The ETF should therefore be paired with a de-risking rule: move one-third of accumulated corpus into short-duration debt once the child is within three years of college entry.

4.4.1 FY2019: NBFC Liquidity Crisis

The IL&FS default in September 2018 triggered a systemic NBFC liquidity crunch. Education NBFCs and consumer finance companies in the basket fell 20–35%. Simultaneously, Education-CPI continued rising at ~10.9%. This is the fundamental hedge limitation of an equity basket: equities carry systemic financial market risk that is orthogonal to the inflation being hedged. Mitigation: the ETF wrapper allows holding through the trough — investors with long horizons (5+ years for an education goal) can absorb this drawdown.

4.4.2 FY2020: COVID-19 and Fee Waivers

The March 2020 crash took the basket down significantly. Education-CPI growth compressed to 7.3% as institutions waived hostel fees and coaching shifted online at discounted rates. The basket recovered fully by FY2021 (+44.5%). Critically, an investor who held through both years still achieved their Education-CPI hedge objective over the full decade. This underscores the importance of communicating minimum 5-year holding periods for this product.

4.5 Comparison: What If the Family Had Used This ETF?

Table 10: Comparative wealth generation across saving instruments for education goals. Education cost benchmark: ₹10 lakh growing at Education-CPI ~10.9% p.a. Simulated; past results not indicative of future performance.

Savings Vehicle (₹10 lakh, FY2015–FY2025)	Corpus (FY2025)	Education Cost (FY2025)	Real Surplus / Deficit
Fixed Deposit (6.5% p.a.)	₹18.7 lakh	₹28.1 lakh	–₹9.4 lakh ✗
Nifty 50 Index Fund	₹26.9 lakh	₹28.1 lakh	–₹1.2 lakh ✗
PPF (7.1% p.a.)	₹20.1 lakh	₹28.1 lakh	–₹8.0 lakh ✗
Education-CPI ETF (simulated)	₹41.0 lakh	₹28.1 lakh	+₹12.9 lakh ✓✓

5. ETF Wrapper Economics & SEBI Positioning

5.1 SEBI Scheme Categorisation

The legal wrapper is a passive equity ETF tracking a custom index. Under SEBI's mutual-fund scheme categorisation, it belongs in "Other Schemes — Index Funds/ETFs". The index must be independently maintained, rule-based, publicly disclosed, and capable of daily portfolio transparency and indicative NAV calculation.

Table 11: SEBI regulatory positioning for the Education-CPI ETF.

Item	Proposed Design
Scheme name	Bharat Shiksha CPI ETF
Benchmark	Bharat Education-CPI Equity Hedge Index
Scheme category	Index Fund/ETF under SEBI mutual-fund categorisation
Replication style	Full replication for liquid large/mid constituents; stratified sampling for capped micro-pure plays if liquidity fails
Creation unit	50,000 ETF units, in-kind basket plus cash component for odd lots, dividends and corporate actions

Rebalance	Semi-annual index rebalance in March and September; monthly Education-CPI publication; extraordinary corporate-action reviews
Disclosure	Daily portfolio, iNAV, tracking difference, tracking error, component weights and Education-CPI methodology note
Investor suitability	Parents and goal planners with seven-plus-year horizon; unsuitable for fee payments due within three years unless paired with debt

5.2 Total Expense Ratio (TER) and Break-Even Analysis

A realistic starting TER is 0.40% to 0.45%. That is higher than plain Nifty/Sensex ETFs because the index is custom, needs education-price maintenance and has more market-making work. It is still low enough to fit passive-product expectations.

Table 12: ETF operating cost structure and break-even analysis.

Cost Line	Annual Estimate	Notes
Index calculation and IP licence	₹25 lakh	Custom index publication, methodology governance and data quality checks
Market-making and exchange listing support	₹30 lakh	Spreads matter more because some pure education names are smaller
Trustee, audit, custody, RTA, legal and compliance	₹45 lakh	Normal ETF operating stack
Research/data maintenance	₹20 lakh	Education-CPI component collection and source audit
Marketing and investor education	₹10 lakh	Goal-planning communication, SIP calculators and risk disclosure
Fixed-cost total	₹1.30 crore	Rounded annual steady-state estimate
Break-even AUM	~₹520 crore	Assumes 25 bps net AMC revenue after variable costs and distribution support

5.3 Creation / Redemption Mechanism

The ETF uses a standard in-kind creation/redemption model via Authorised Participants (APs), consistent with all NSE-listed ETFs:

- Creation Unit size: 50,000 units, in-kind basket plus cash component.
- Basket file published daily at 9:00 AM by the AMC.
- Cash component permitted up to 5% of basket value for fractional share handling.
- T+1 settlement for primary market; T+2 for exchange secondary market.
- Minimum 2 APs committed at launch; spread requirement of $\leq 0.50\%$ iNAV during market hours.

5.4 Index IP and Licensing

The ETF sponsor can own the Education-CPI methodology while outsourcing calculation to an independent index administrator. That creates a defensible product moat without converting the fund into active management. The rules, weights, caps, rebalance calendar and data hierarchy remain public.

5.5 Go-to-Market and Distribution Strategy

The Education-CPI ETF's primary distribution channel is financial advisors and mutual fund distributors (MFDs) who serve goal-based planning clients. The pitch should avoid overpromising. "Beat education inflation" is too dangerous. A better line is: "A transparent equity sleeve built for child-education inflation, to be used with SIPs and a debt glide path." Secondary channels include:

- **Direct-to-parent digital:** Partner with coaching platforms (Allen, FIITJEE, Vedantu) to offer SIP integration at fee payment touchpoints — parents already thinking about education costs are pre-qualified investors.
- **Corporate benefits:** HR platforms can offer Education-CPI SIPs as a child education benefit, analogous to NPS for retirement.
- **Jan Dhan Linkage:** A co-investment product with the government's SSDP (Sukanya Samridhi) ecosystem to address the girl-child education savings gap.
- **IFA network:** AMFI-registered advisors earn 50 bps trail on assets — standard for equity ETFs.

The marketing narrative is simple: 'Your FD earns 6.5%. Education costs rise at 11%. Invest in what makes education expensive — and let inflation work for you, not against you.'

6. Risk Factors, Limitations & Conclusion

6.1 Key Risk Factors

Market Risk

The ETF invests in listed equities. All basket stocks carry standard market risk, including sector-specific downturns (NBFC liquidity crises), company-specific events, and macroeconomic shocks. The ETF is not capital-protected. Maximum drawdown in back-test: -24.1% (FY2019–FY2020).

Hedge Imperfection Risk

The basket's annual R^2 to Education-CPI is 0.01 — meaning that in any single year, equity markets may fall significantly while Education-CPI continues to rise. However, the three-year rolling beta of 2.52 and cumulative coverage ratio of 1.46x confirm that the hedge works over medium-to-long horizons. Investors must hold for minimum 5 years to capture the mean-reversion of the hedge relationship.

Index Concentration Risk

The basket has meaningful concentration in financial services companies (financing rails at 45%). If SEBI introduces new regulations on education loan interest rates (a public policy risk, given India's political sensitivity around education costs), the basket could be materially impacted.

Liquidity Risk

Several basket constituents (CL Educate, Career Point, Kokuyo Camlin) have relatively thin daily trading volumes. At AUM below ₹200 crore, the ETF may face market-impact costs on rebalancing that erode returns.

Regulatory Risk

India's private education sector is subject to state-level fee regulation (fee regulatory committees in states like Maharashtra, Tamil Nadu cap private college tuition). If fee regulation tightens, the Education-CPI itself would fall — a positive outcome for families but a negative for the ETF basket's revenue-inflation linkage.

6.2 Limitations of This Study

- The Education-CPI series is reconstructed, not officially compiled. While grounded in MoSPI data, AISHE, and public fee surveys, the composite is estimated and cannot be verified with the precision of an official statistical release.
- The back-test is simulated and subject to survivorship bias, data availability limitations, and the absence of actual market microstructure constraints.
- Several basket constituents were unlisted or thinly traded in 2015; their inclusion in historical back-tests uses proxy instruments, introducing additional estimation error.
- The Education-CPI inflation rate of ~10.9% is a constructed estimate. The official MoSPI CPI Education sub-index shows lower rates (~6–7%) because it captures a broader and lower-inflation slice of the education sector.

6.3 Why This Product Has Not Existed Yet

The white space exists for four reasons. First, official CPI is not wrong, but it is not designed for household goals. Second, education prices are fragmented across public institutes, private colleges, coaching centres and city living costs. Third, India's listed education universe is shallow, so a pure-play ETF would struggle with capacity and liquidity. Fourth, "education inflation hedge" is a sensitive phrase: if marketed carelessly, investors may mistake it for a guarantee.

The solution is to be explicit about the compromise: the ETF is a transparent, rules-based equity hedge for the long accumulation phase. It measures the right liability, buys companies with plausible revenue pass-through, and admits that the final three years before college need a glide path into lower-risk assets.

6.4 Conclusion

India has 1.4 crore families saving for a child's higher education with no instrument specifically designed to hedge the inflation they face. The gap between headline CPI (~5.1% CAGR) and true education inflation (~10.9% CAGR) compounded over a decade produces a corpus shortfall of 30–40% for most conventional saving strategies. The Education-CPI ETF fills this white space with a rules-based, SEBI-compliant, listed equity product whose constituent revenues structurally rise with education inflation.

The back-test demonstrates that over FY2015–FY2025, the proposed basket delivered a CAGR of 15.2% versus Education-CPI's 10.9%, achieving a cumulative coverage ratio of 1.46x. The hedge broke down during systemic equity market shocks (FY2019 NBFC crisis, FY2020 COVID) — a risk that is mitigated by the long investment horizon of education savers and the recommended three-year glide path into debt.

The Bharat Inflation Wallet — of which the Education-CPI ETF is the first and most analytically robust component — represents a genuine productisation of goal-based financial planning. Every certified financial planner in India already teaches clients to think in buckets: education, healthcare, housing,

aspirational. The Education-CPI ETF gives that insight an investable form for the first time. The white space is real. The demand is latent. The methodology is sound. The product is ready to be built.

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Appendix A. Methodology and Data Notes

Base year. The production Education-CPI would be compatible with the current official CPI base series, while this paper rebases all historical series to FY2015 = 100 so that a decade-long back-test can be read on one page.

Education-CPI construction. Tuition, living, coaching, materials and credit costs are indexed separately, then combined using fixed persona weights. Weights are reviewed every two years against education-expenditure surveys and AISHE enrolment composition. One-off institutional fee shocks are included but phased to avoid a single institute creating a false national signal.

ETF back-test. The paper uses a rules-based reconstruction rather than a live index history because the product does not yet exist. Constituents are selected from listed Indian companies with education revenue linkage, sufficient trading history and public disclosures. Where companies listed after FY2015, their weight enters only after eligibility and seasoning. Returns are shown as rounded annual total-return estimates for the basket, net of neither TER nor tax.

Use of evidence. The paper deliberately separates official data from model choices. Official data establish the macro and sector context; the Education-CPI weights and ETF basket are product-design decisions that must be disclosed and governed.
